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# Autonomous Dual Interchangeable Robotic Arms for In-Space Servicing, Assembly, and Manufacturing

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## Background

- The In-Space Servicing, Assembly, and Manufacturing (ISAM) market is experiencing exponential growth, projecting a value of $135.3 billion by 2040[1]
- There is a critical need for new technologies to extend mission lifespans and reduce operational costs
- Existing space robotic systems are too large, expensive, and specialized for the small satellite market in LEO

**Objective of this design:**
Provide a small-scale and low-cost twin robotic arm system architecture with interchangeable tools to enable novel ISAM capabilities.

## System Design

- GATOR Arms is currently designed to be mounted to a BCT X-Sat Venus Class Bus
- Two 6 DOF robotic arms with a total mass of 20.4 kg
- Universal interface to enable end effectors to be interchanged
- Tool belt to store 10 different tools including a gripper, drill, wrench, camera, and welder
- The length of the arms is 30in (76.2cm) with a diameter of 3in (7.62cm)

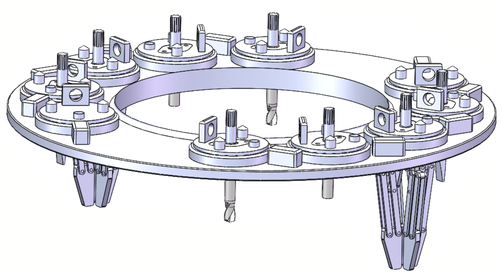

Fig. 1. Tool belt capable of holding 10 unique tools

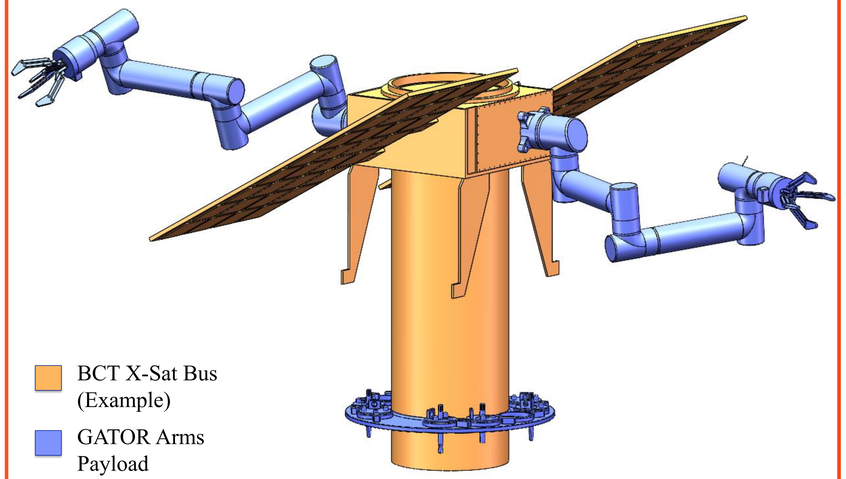


Fig. 2. GATOR Arms on a BCT X-Sat Venus Class Bus

## Interchangeable End Effectors

- Universal interface design allows for any tool to be implemented in the tool belt system
- End-effector system allows quick tool swapping using an interlocking solenoid connection
- Geometric extrusions ensure repeatable, precise alignment for correct tool orientation

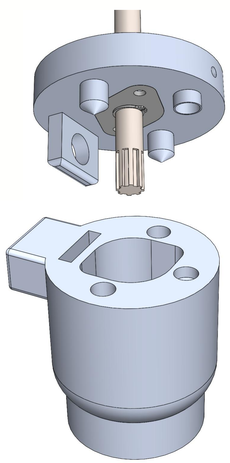

Fig. 3. Interchangeable Interface

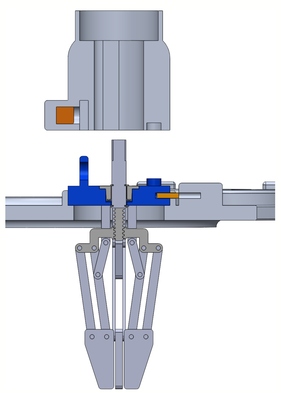

Fig. 4. Section view of the arm changing tools

## Design Justification

**Cost-effective**

- Aluminum Construction
- COTS motors and electronics
- End effector interface
- Initial prototype: < $7,000

**Universal Design**

- Use across different satellites
- Modular interchangeable end effectors
- Service or replace solar panels, propulsion systems, deployables, etc.

## Discussion

**Prototype Development and Future Testing**

- GATOR Arms are currently at TRL 2, with full CAD models and a proof-of-concept model is in progress
- Two prototypes are to be completed by Oct. and Dec. 2025 of exit criteria TRL 4 and 5 to test mechanical and autonomous operations
- A GATOR Arms flight test requires integration on a suitable small-sat in LEO to validate precision, power, cooperation, and modularity

**Technological Challenges and Risk Mitigation**

- Further work needs to be done on the computer vision system, an image data set for identifying damage, and stepper motor precision
- Use of two arms creates redundancy to mitigate risk of motor failure. The arms also use aluminum to lower risk of radiation damage

## Contact

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